# LAB 1 SOLUTION DOCUMENT

**Course:** Cloud Computing (CC-F26)

**Section:** BSE-V(A)

**Instructor:** Prof. Waqas Saleem

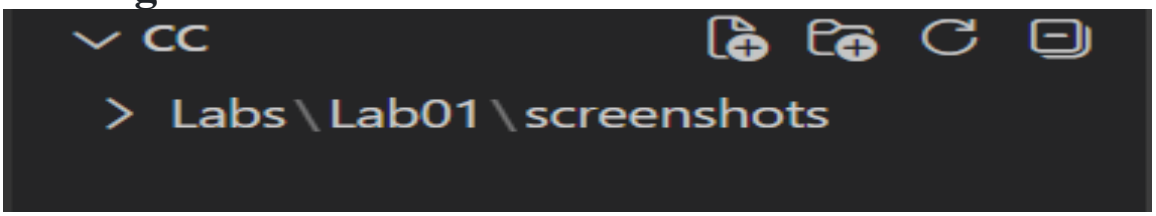
**Date:** September 13, 2026

## STUDENT INFORMATION

- **Student Name:** Kashaf Noor
- **Registration Number:** FA24B1-SE-029
- **Course Code:** CC-F26
- **Section:** BSE-V(A)
- **GitHub Username:** kashafnoor22

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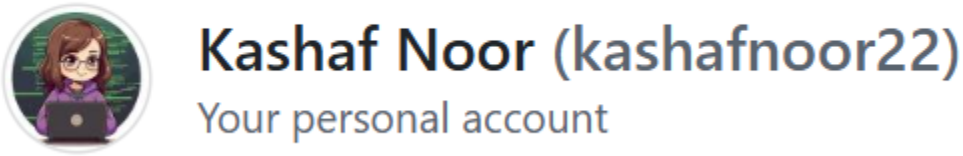
## Getting Started




# Task 1: Create and Configure Your GitHub Account

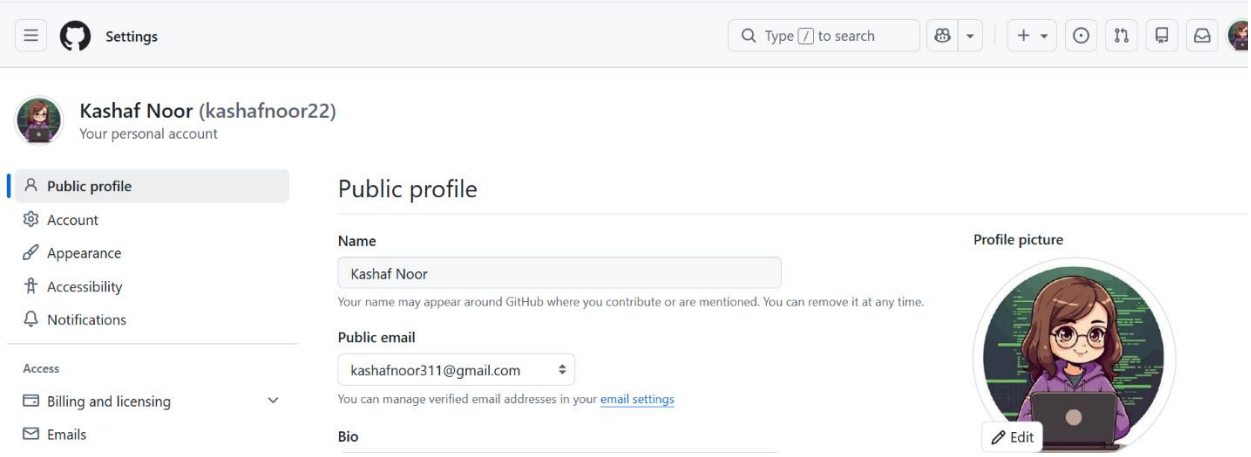
This task verifies the setup of the primary GitHub account used throughout the course.

### Evidence 1.1: GitHub Profile




Caption: Screenshot showing public GitHub profile for user laibaamjad065.

### Evidence 1.2: GitHub Actual Name Field

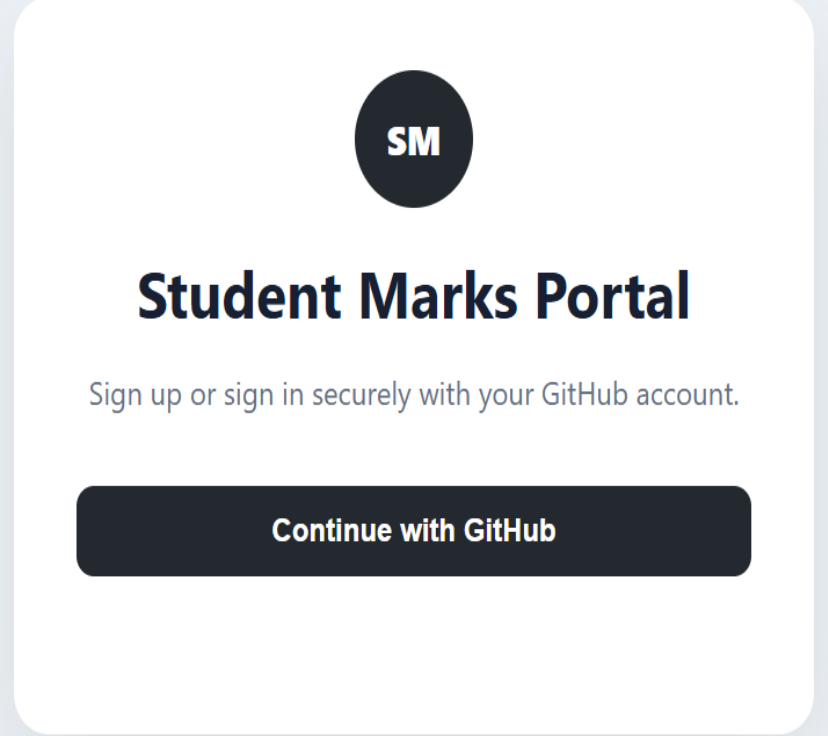



Caption: GitHub public settings page displaying complete actual name 'Laiba Amjad'.

# Task 2: Register in the Student Marks Portal

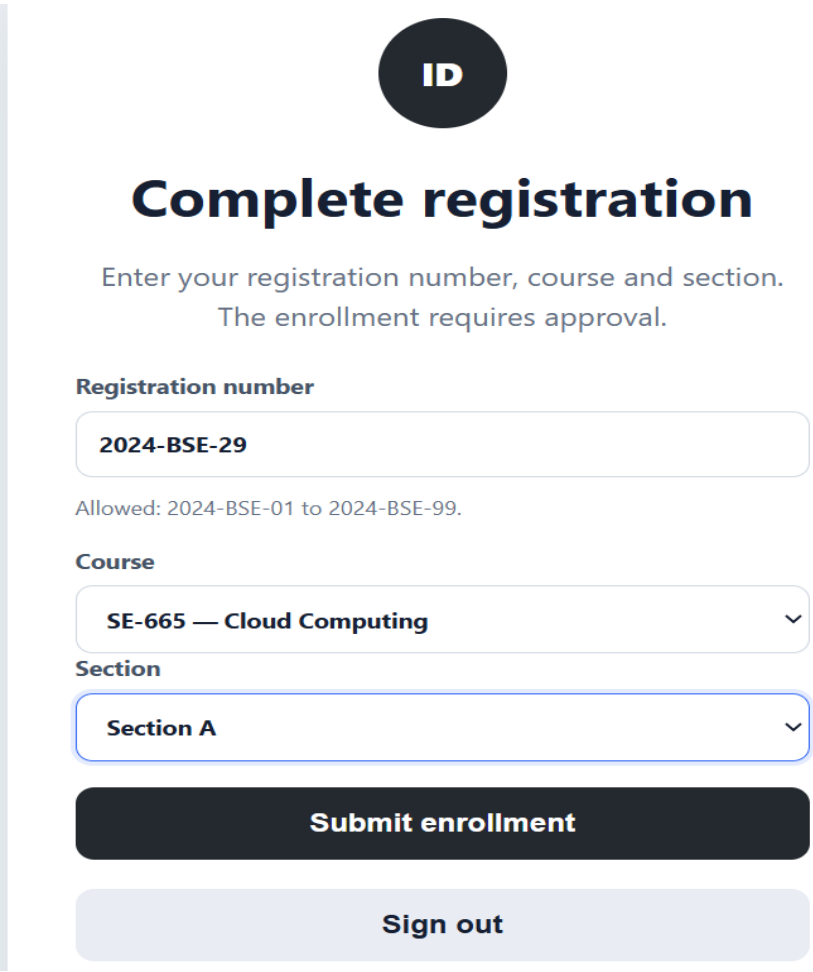
This task documents registration on the official Student Marks Portal using GitHub OAuth.

## Evidence 2.1: Portal Login Page



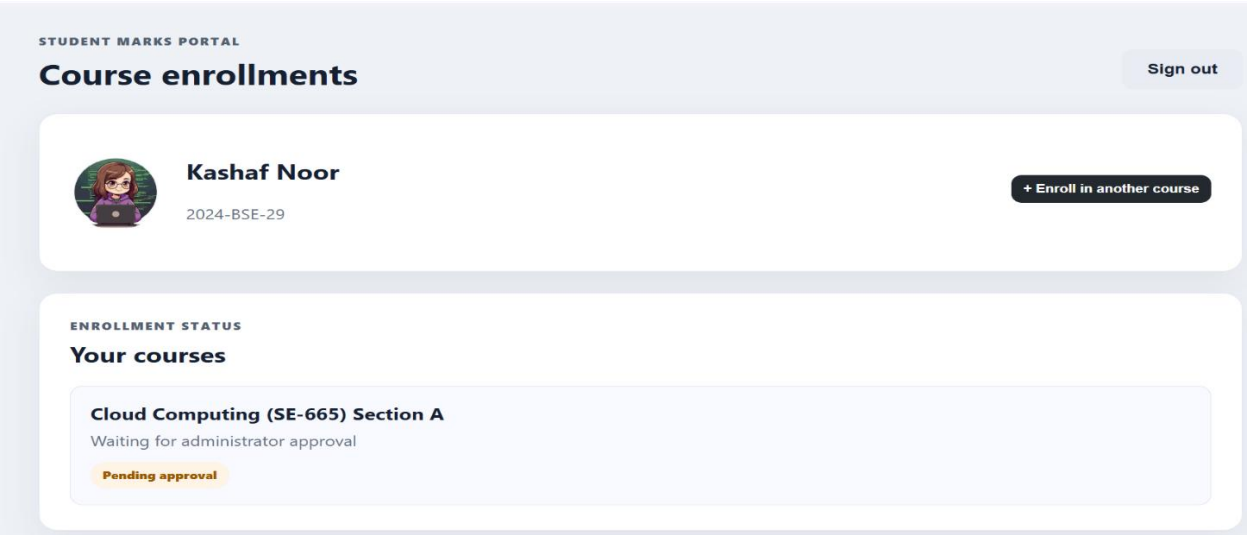

Caption: Initial landing page of the Student Marks Portal showing GitHub Sign-In option.

## Evidence 2.2: Portal GitHub Profile Link




Caption: Successful authentication into portal displaying linked GitHub account credentials.

## Evidence 2.3: Enrollment Submission Confirmation

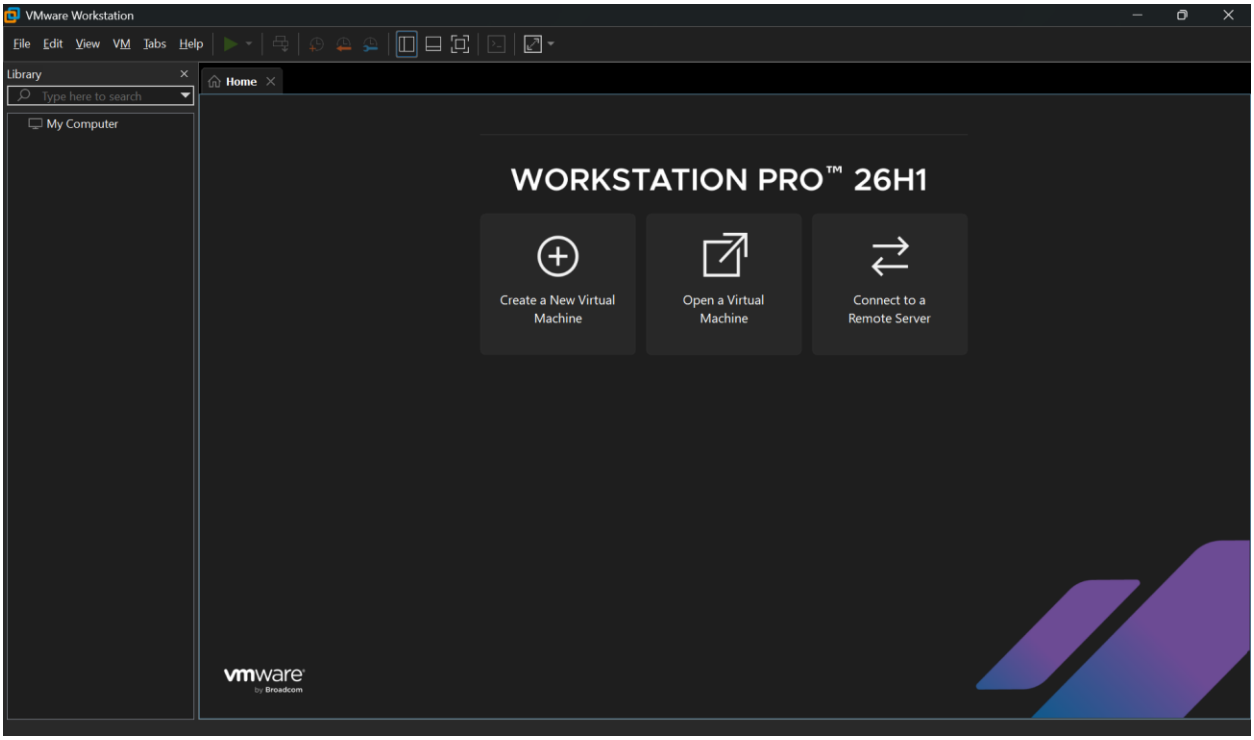


Caption: Portal dashboard confirming course enrollment submission for CC-F26, Section BSE-V(A).

# Task 3: Download VMware Workstation Pro and Ubuntu Server

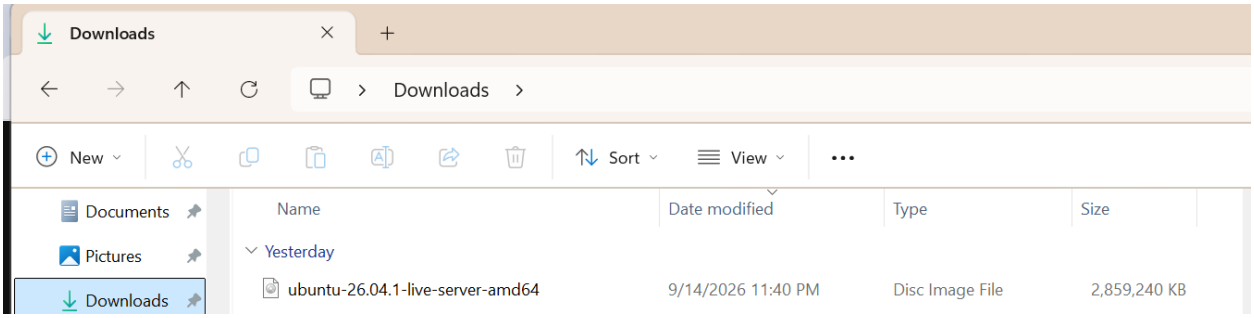
Verification of hypervisor installation, ISO download, and host machine disk capacity.

## Evidence 3.1: VMware Workstation Pro Status



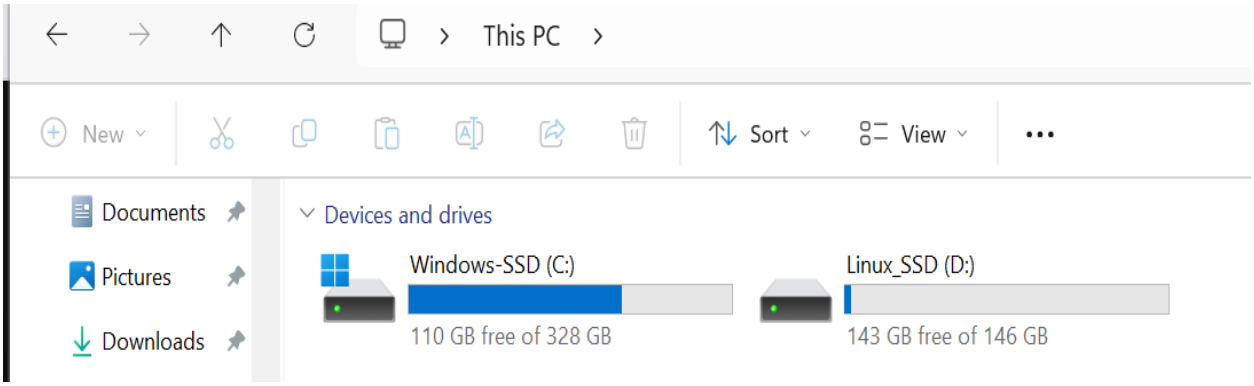

Caption: VMware Workstation Pro running on Windows host machine.

## Evidence 3.2: Ubuntu Server ISO Verification




Caption: Downloaded Ubuntu Server 24.04 LTS ISO file in local storage directory.

## Evidence 3.3: Storage Availability

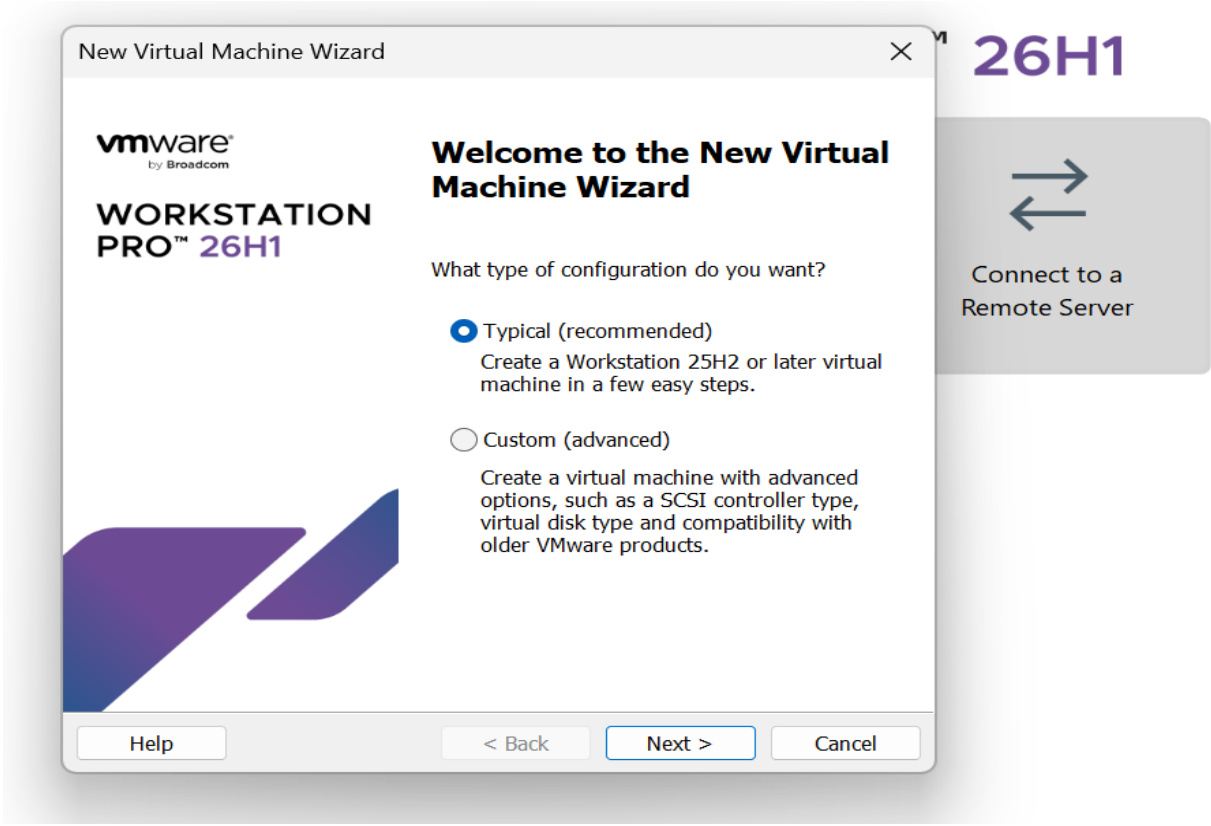



Caption: Host storage drive showing sufficient free disk space allocated for VM creation.

# Task 4: Create the Ubuntu Server Virtual Machine

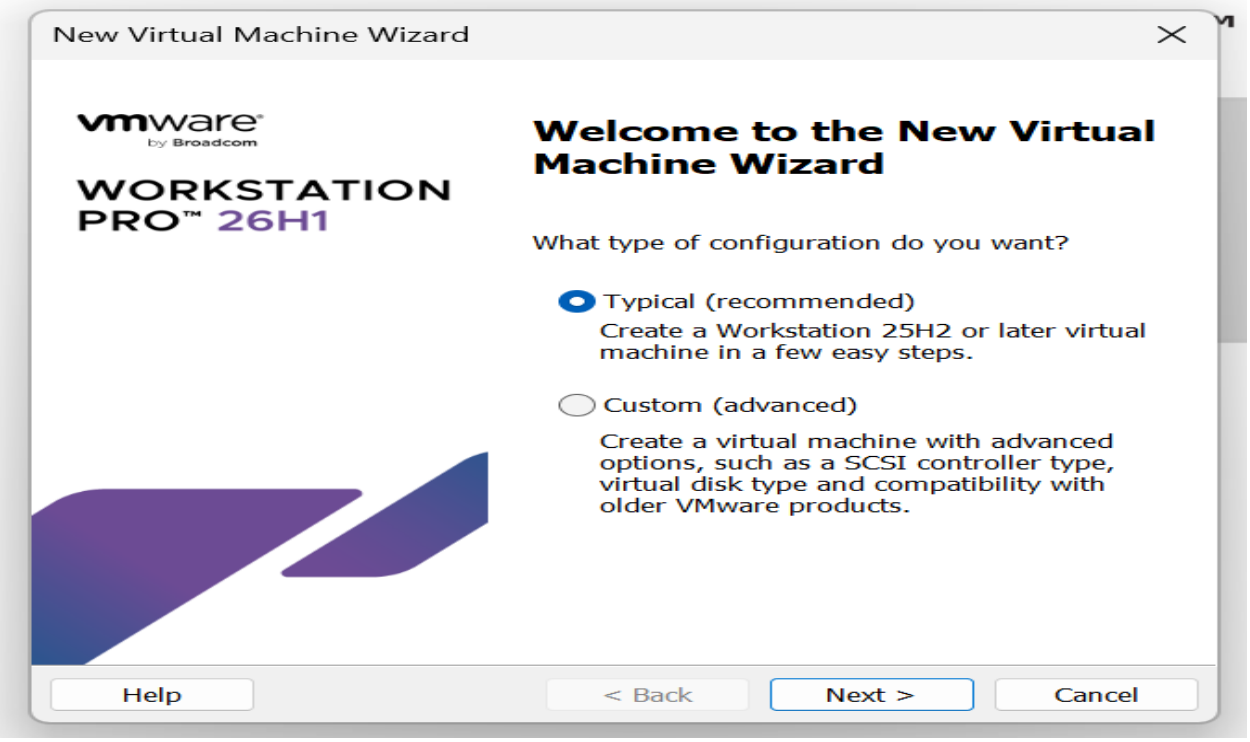
Step-by-step setup of the Virtual Machine hardware parameters within VMware.

## Evidence 4.1: VM Creation Wizard



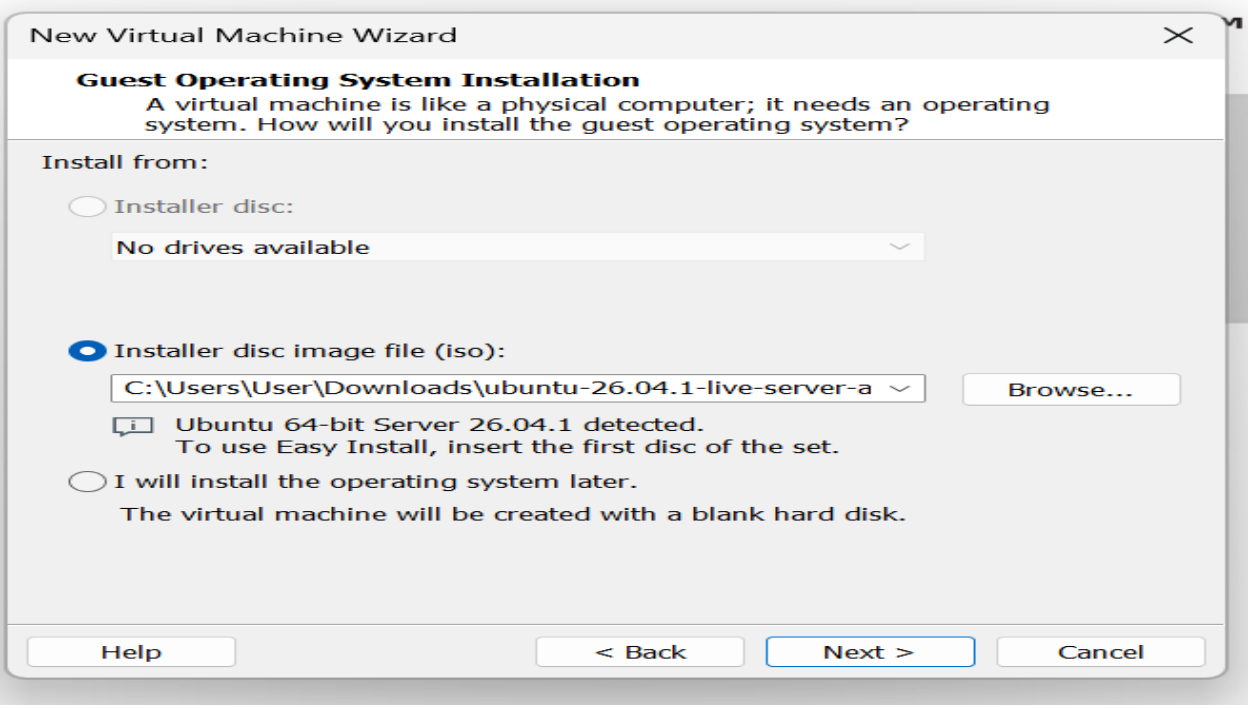

Caption: Opening screen of VMware Virtual Machine Creation Wizard.

## Evidence 4.2: Typical Configuration Selection



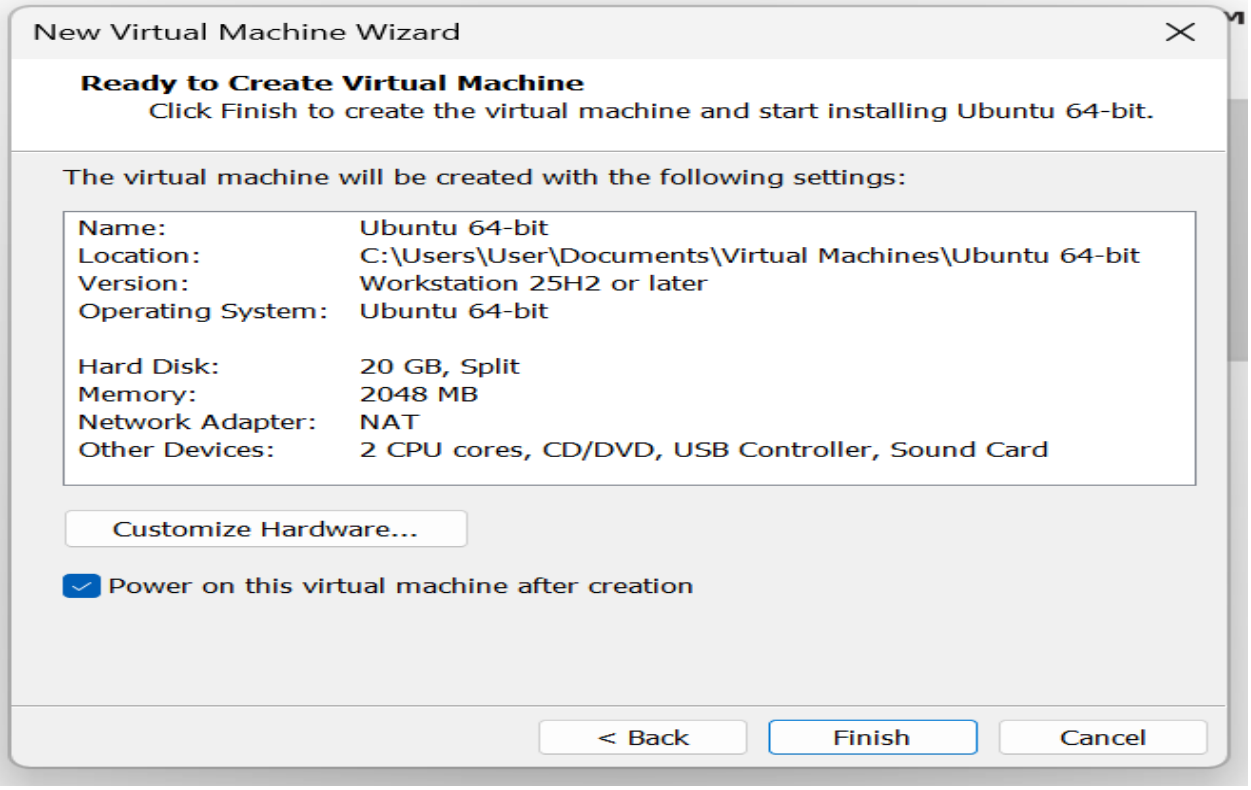

Caption: Selection of Typical (recommended) VM configuration option.

## Evidence 4.3: ISO Mount Selection



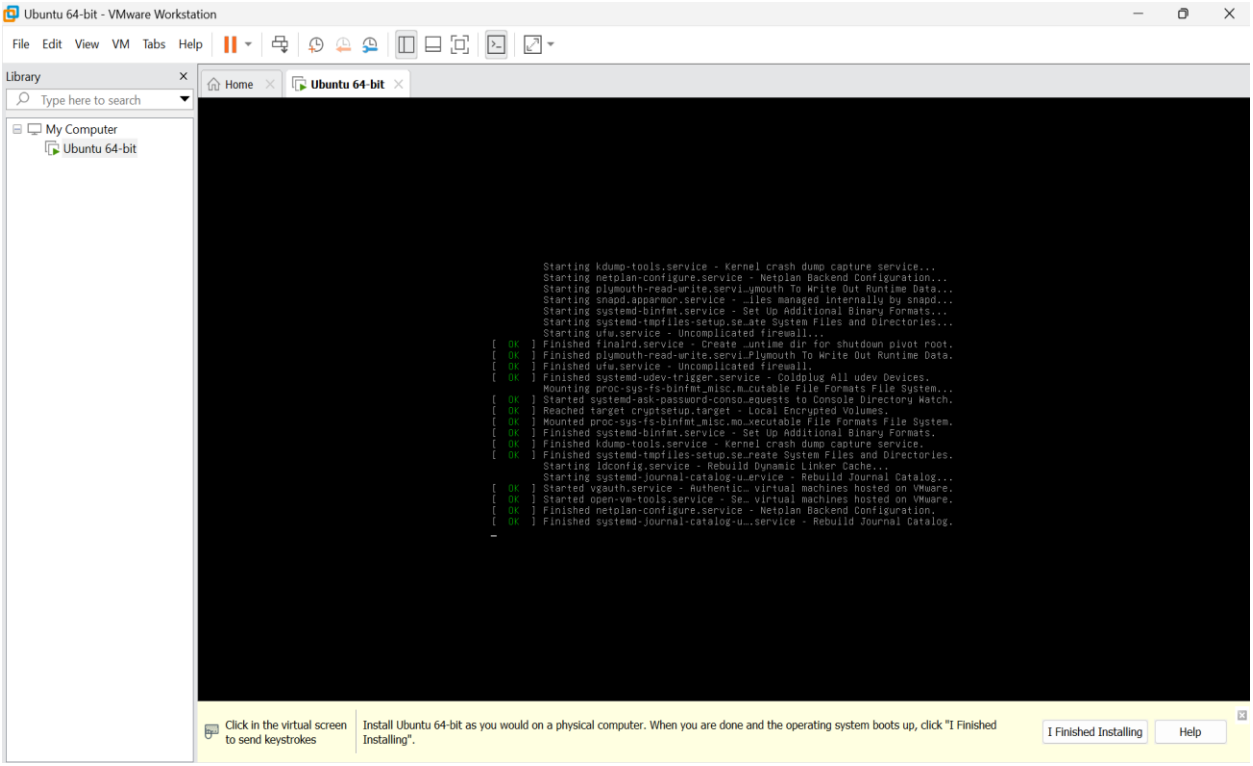

Caption: ISO path pointing to the downloaded Ubuntu Server image.

## Evidence 4.4: VM Hardware Summary




Caption: Final configuration summary window showing CPU, RAM, and Disk allocations.

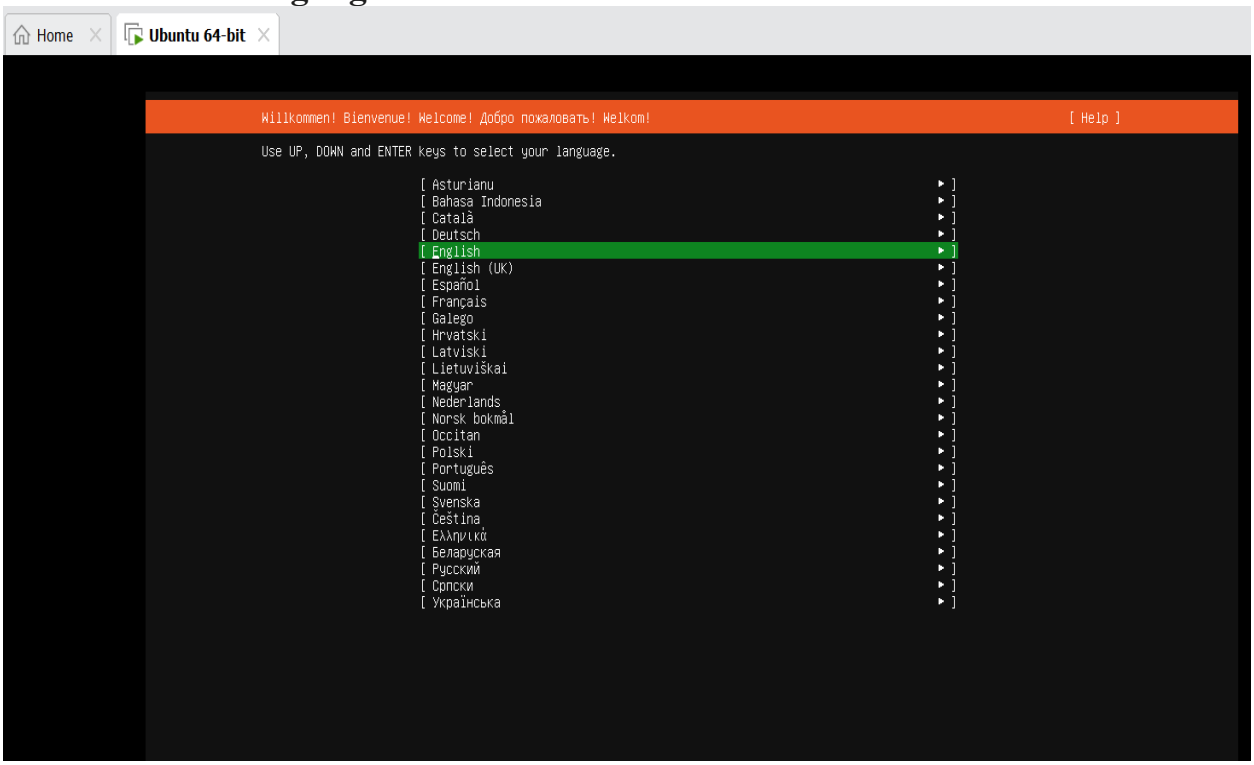
## Evidence 4.5: Initial Boot Screen



Caption: Ubuntu Server GRUB boot loader screen upon initial VM power-on.

# Task 5: Choose the Installation Language

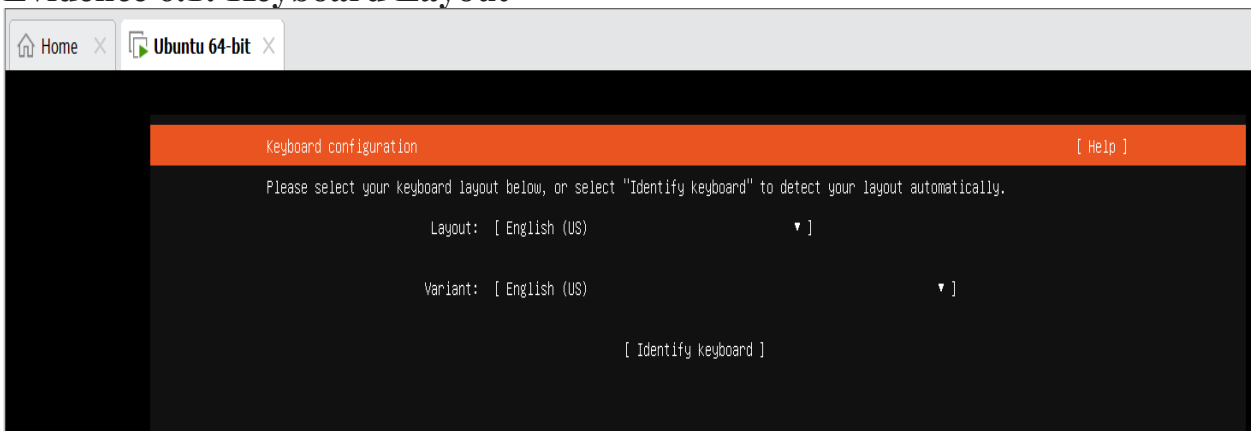
## Evidence 5.1: Language Selection




Caption: Selecting English as the default installation language for Ubuntu Server.

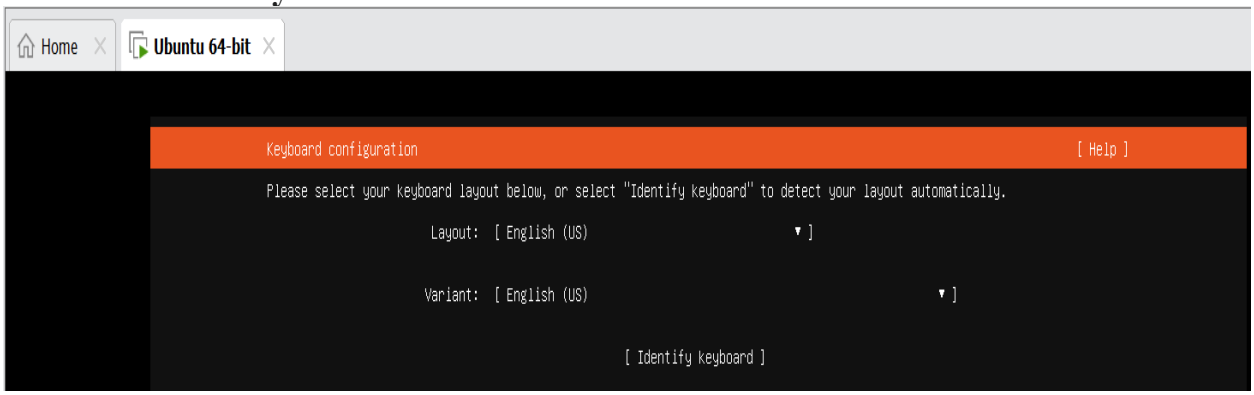
# Task 6: Configure the Keyboard Layout

## Evidence 6.1: Keyboard Layout




Caption: Setting English (US) as primary keyboard layout.

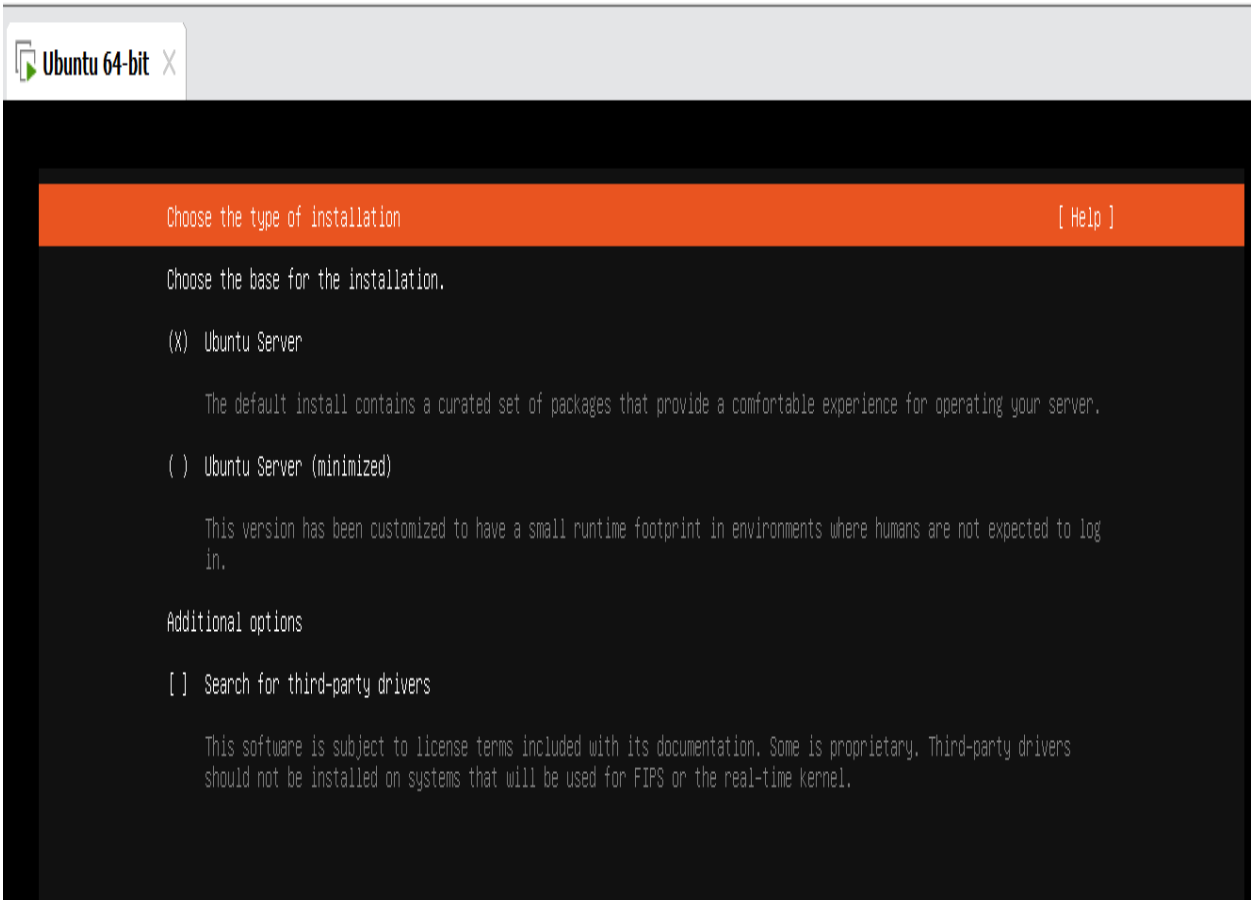
### Evidence 6.2: Keyboard Variant




Caption: Confirming English (US) variant configuration.

## Task 7: Choose the Ubuntu Server Installation Type

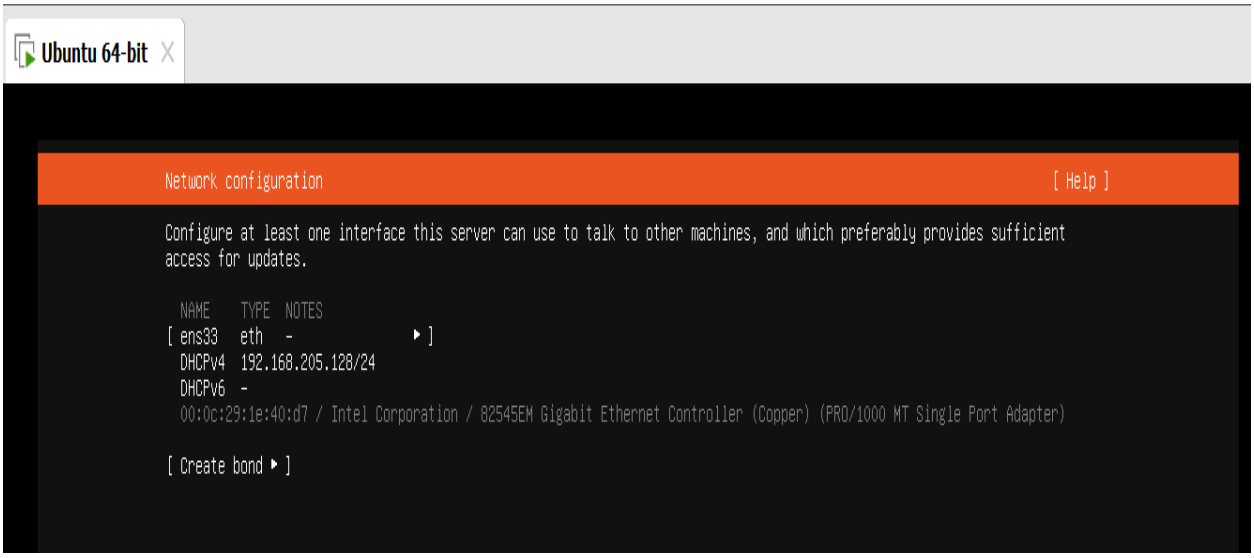
### Evidence 7.1: Installation Type




Caption: Selecting standard Ubuntu Server installation profile.

## Task 8: Configure the Network

### Evidence 8.1: Network Interface Detection




Caption: Installer detecting primary network interface ens33.

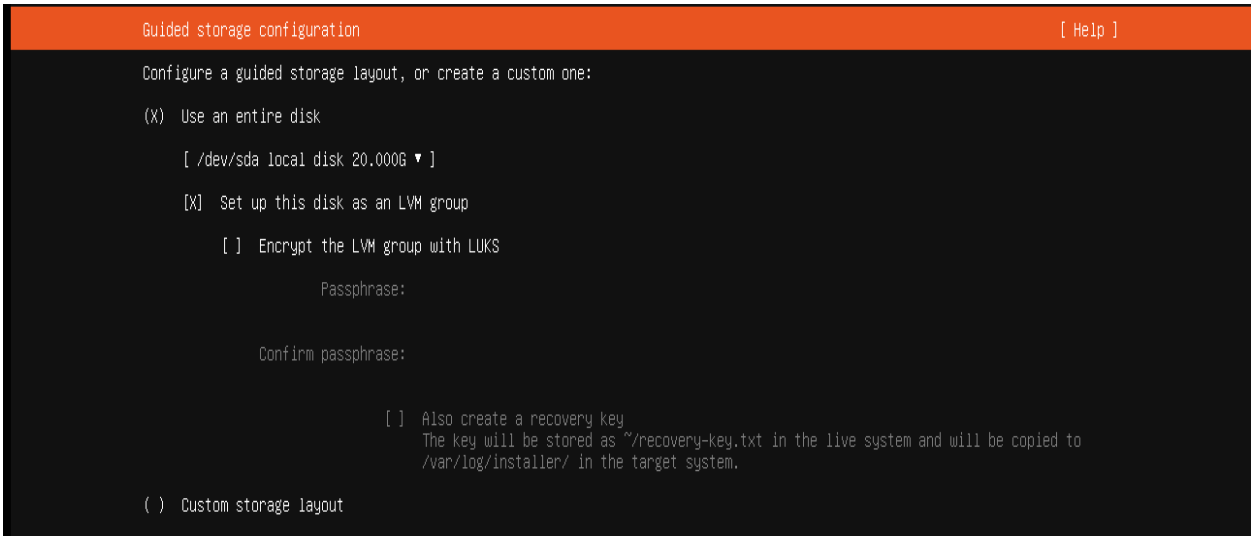
### Evidence 8.2: DHCP Address Assignment




Caption: Successful automatic IPv4 address assignment via DHCP.

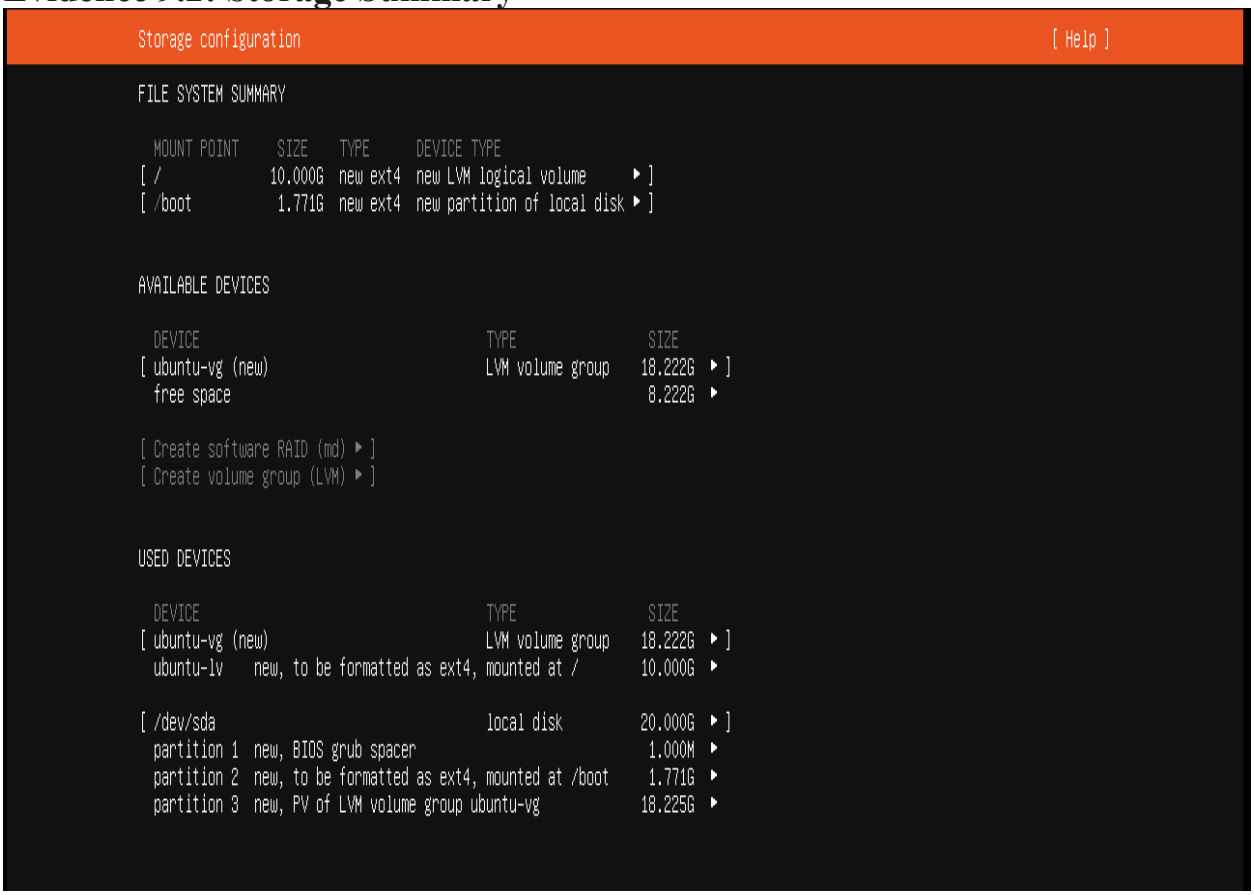
## Task 9: Configure Storage

### Evidence 9.1: Guided Storage Setup




Caption: Choosing guided storage layout utilizing entire virtual disk.

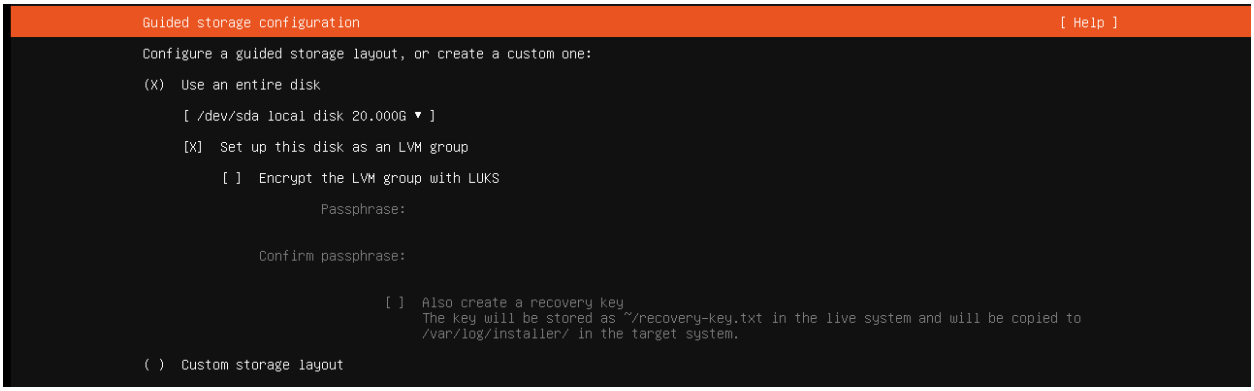
## Evidence 9.2: Storage Summary




Caption: Reviewing overall storage layout before disk allocation.

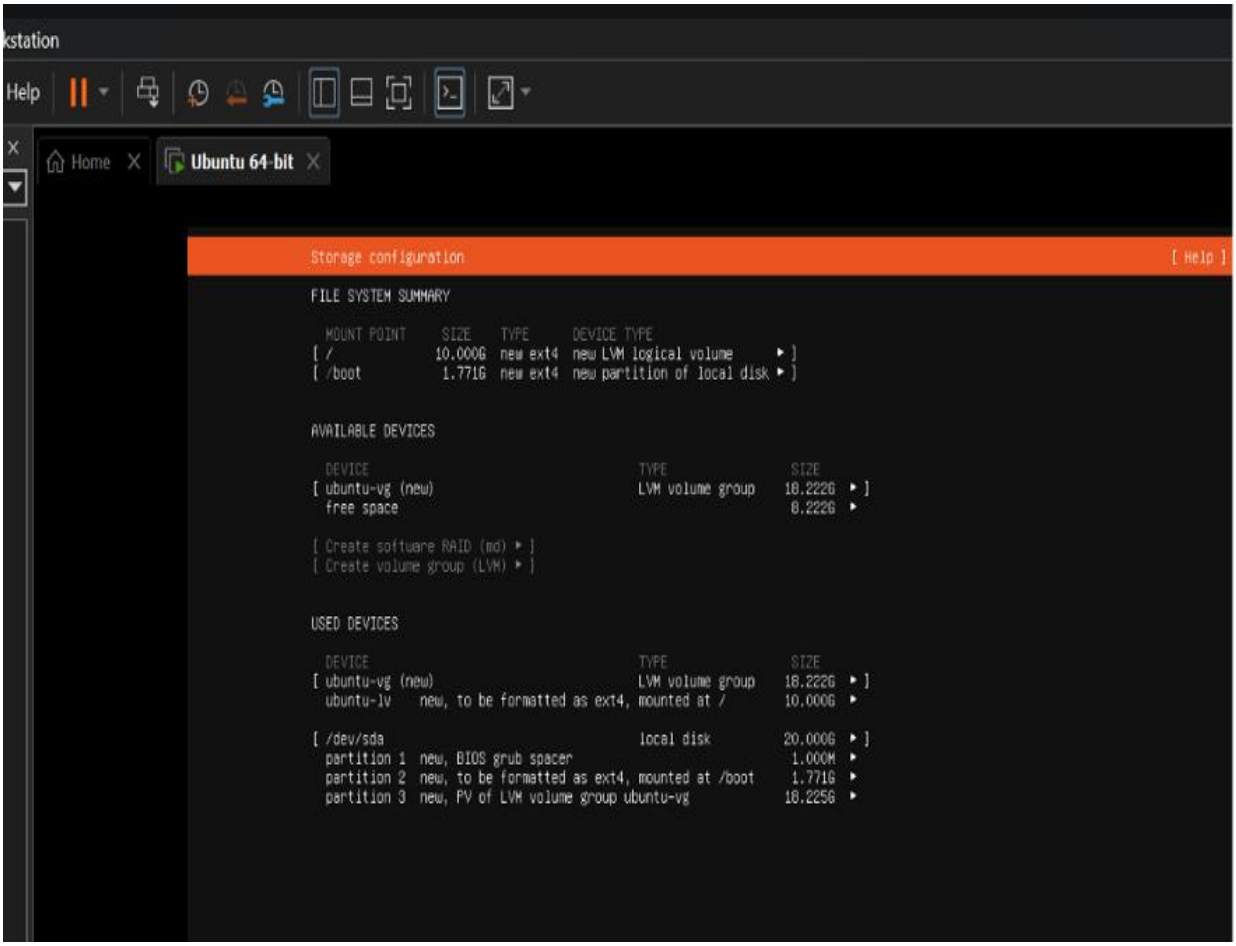
# Task 10: Select the Installation Disk

## Evidence 10.1: Disk Selection




Caption: Selecting target VMware Virtual Disk for installation.

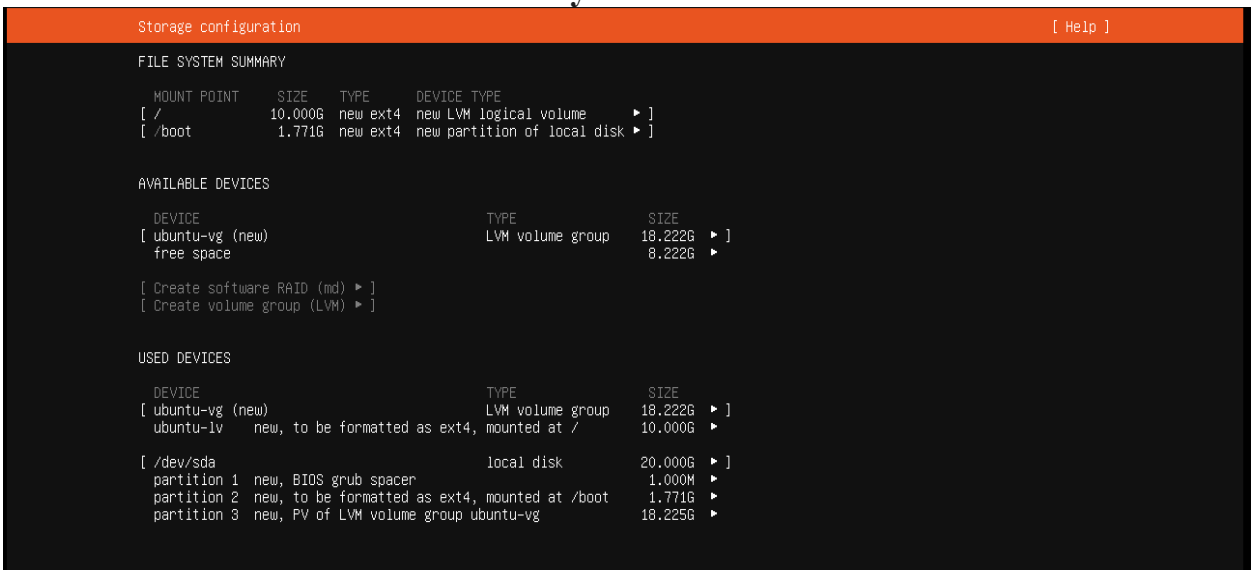
## Evidence 10.2: Disk Size Verification




Caption: Verifying virtual disk size matches allocated capacity (20GB+).

# Task 11: Review the Partitions

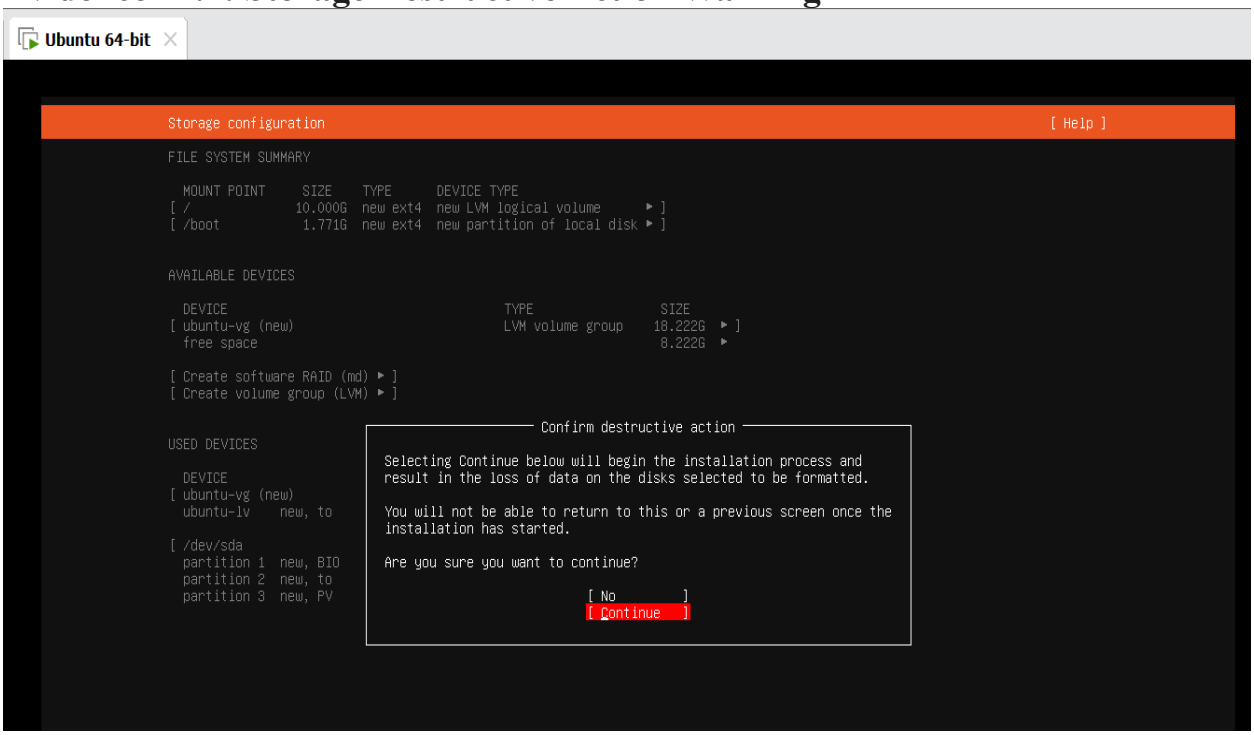
## Evidence 11.1: Partition Scheme Layout




Caption: Detailed view of root (/) and boot partition assignments.

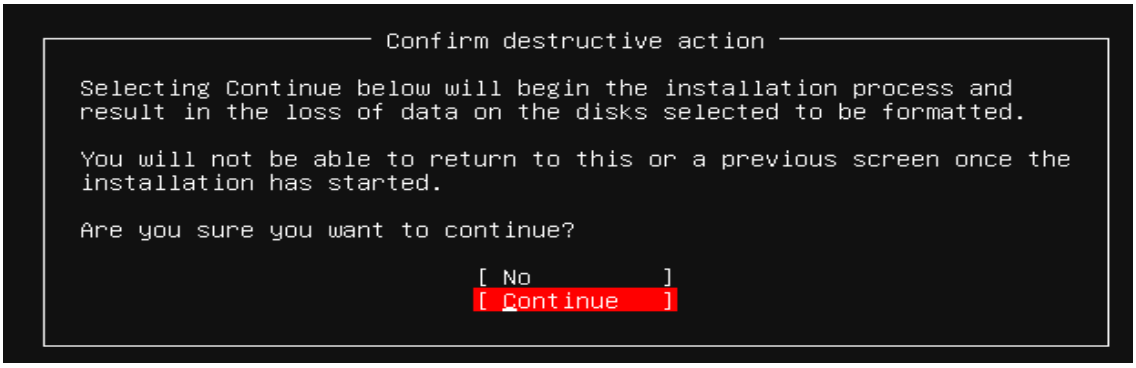
# Task 12: Confirm the Storage Changes

## Evidence 12.1: Storage Destructive Action Warning




Caption: Installer confirmation dialog warning about writing changes to disk.

### Evidence 12.2: Execution Confirmation

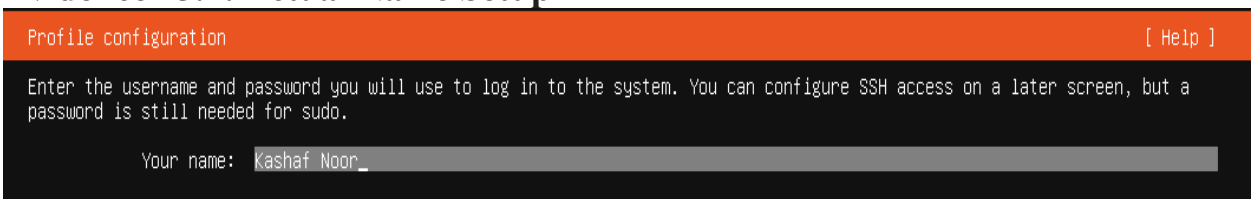



Caption: Selecting 'Continue' to initiate disk formatting and partitioning.

## Task 13: Configure the Ubuntu Profile

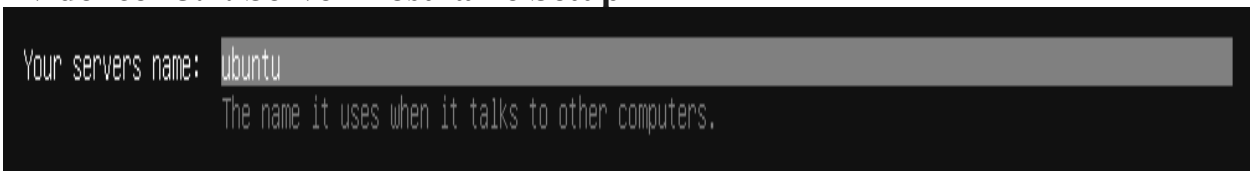
Setting up system administrative account identity.

### Evidence 13.1: Actual Name Setup




Caption: Profile configuration screen entering 'Laiba Amjad' in Your Name field.

### Evidence 13.2: Server Hostname Setup




Caption: Setting system hostname as ubuntu.

### Evidence 13.3: System Username Setup




Caption: Setting system username to match GitHub username laibaamjad065.

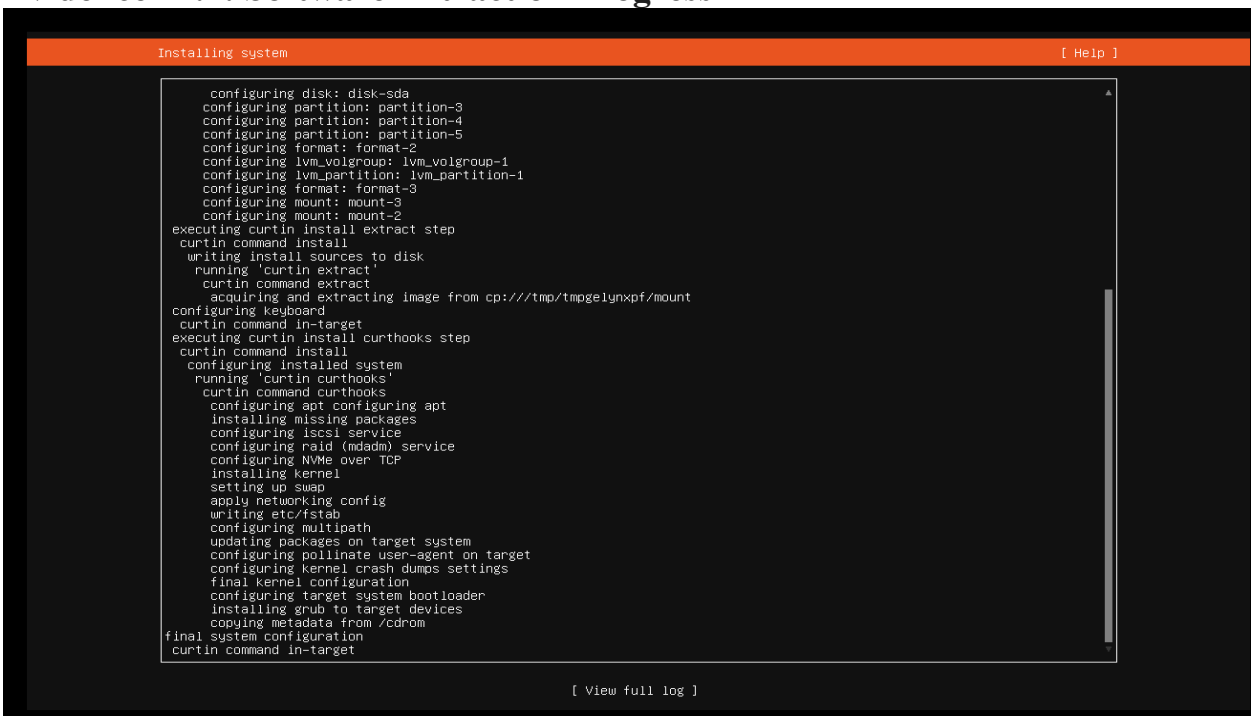
### Evidence 13.4: Completed Profile Summary




Caption: Complete user profile setup screen prior to system installation.

# Task 14: Complete the Software Installation

### Evidence 14.1: Software Extraction Progress

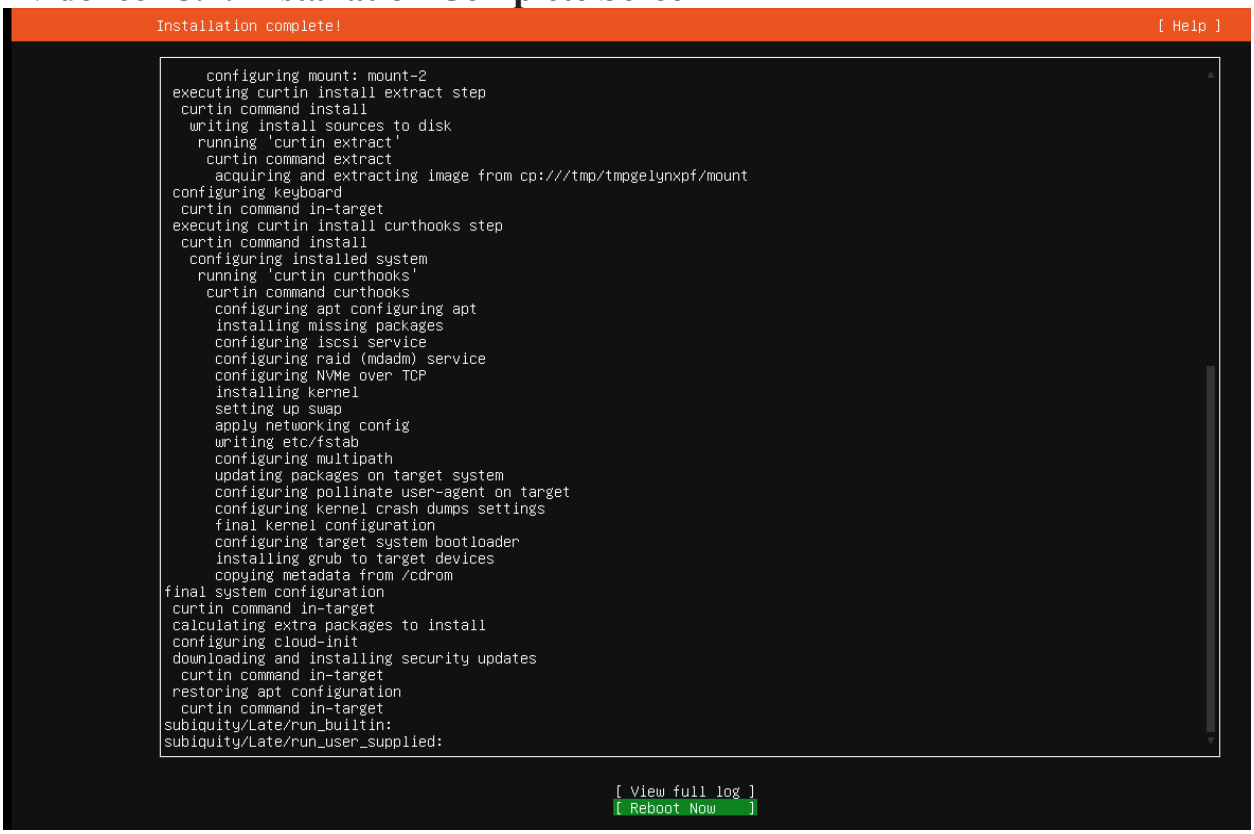



Caption: Core system packages unpacking and installation progress bar.

# Task 15: Reboot and Verify the Ubuntu Identity

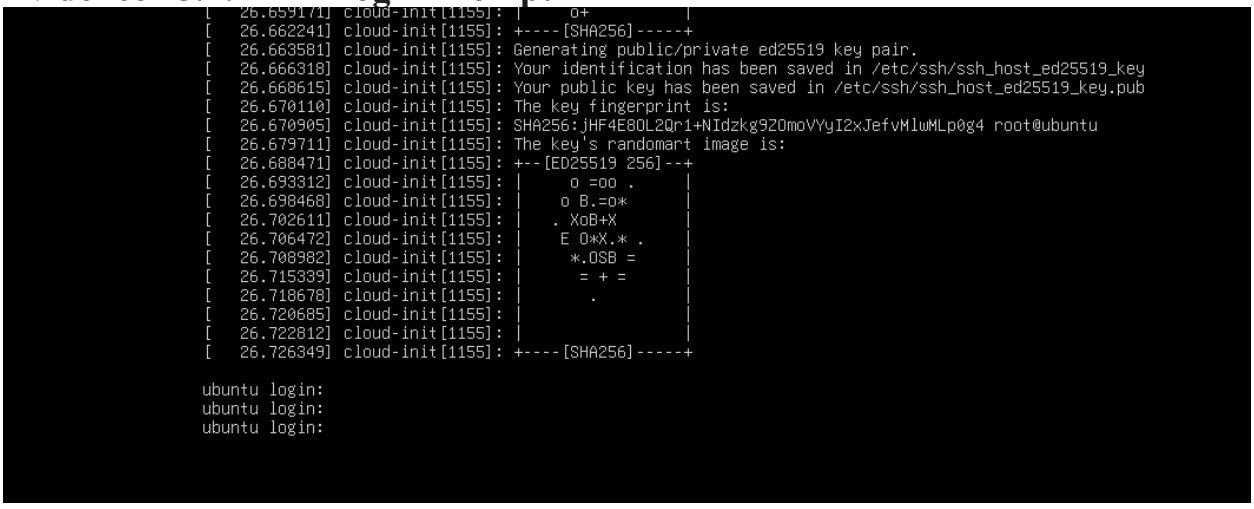
System installation finalization, rebooting, and initial shell authentication.

## Evidence 15.1: Installation Complete Screen



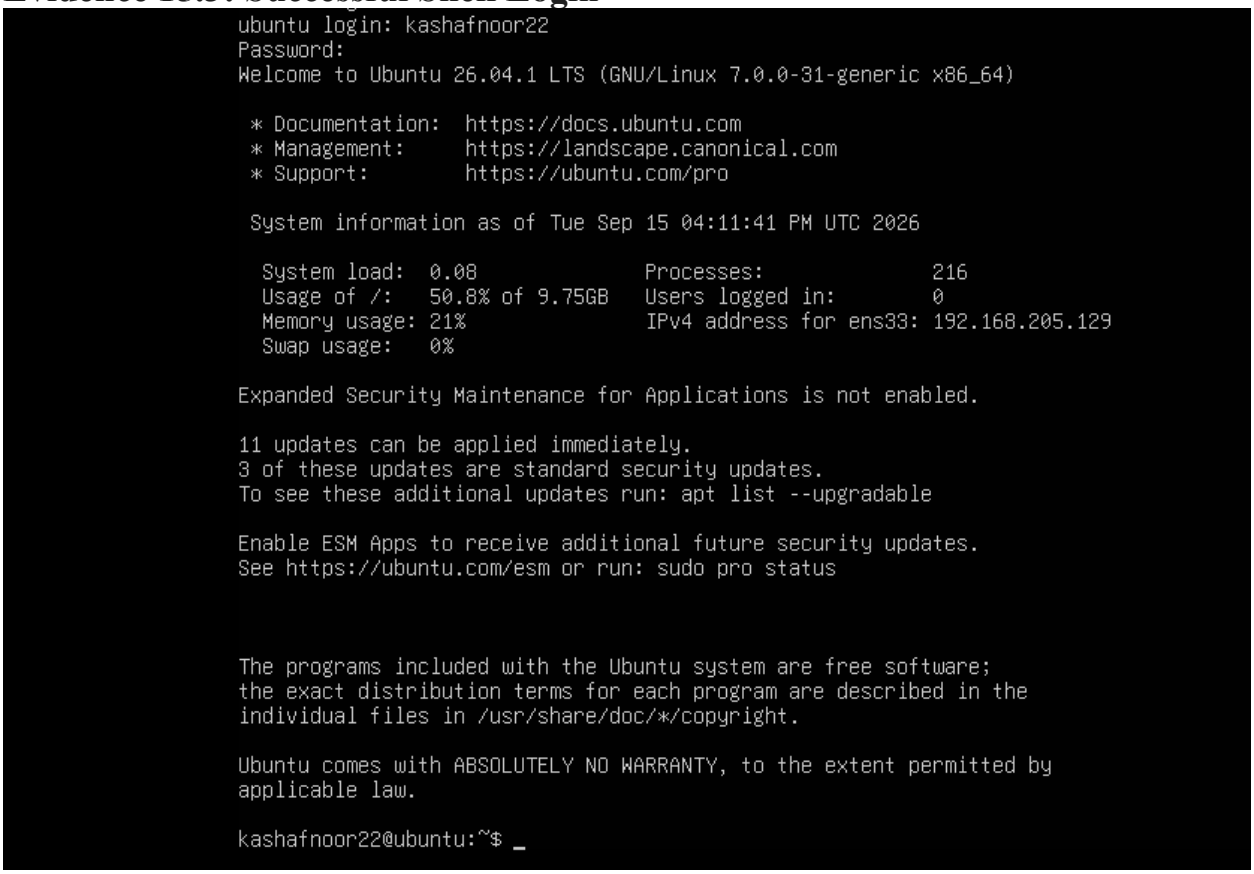

Caption: Installer screen confirming successful software setup and prompt to reboot.

## Evidence 15.2: TTY Login Prompt




Caption: System startup completing at the ubuntu login: prompt.

### Evidence 15.3: Successful Shell Login




Caption: Welcome banner (MOTD) displayed after successful password authentication.

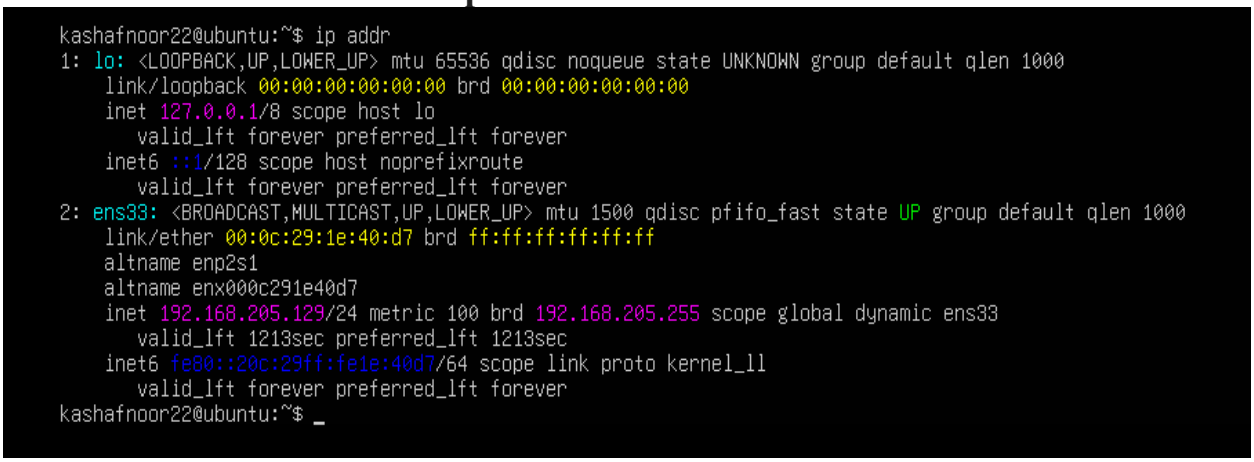
### Evidence 15.4: Terminal Identity Verification




Caption: Active Bash terminal prompt displaying identity kashafnoor22@ubuntu:~$.

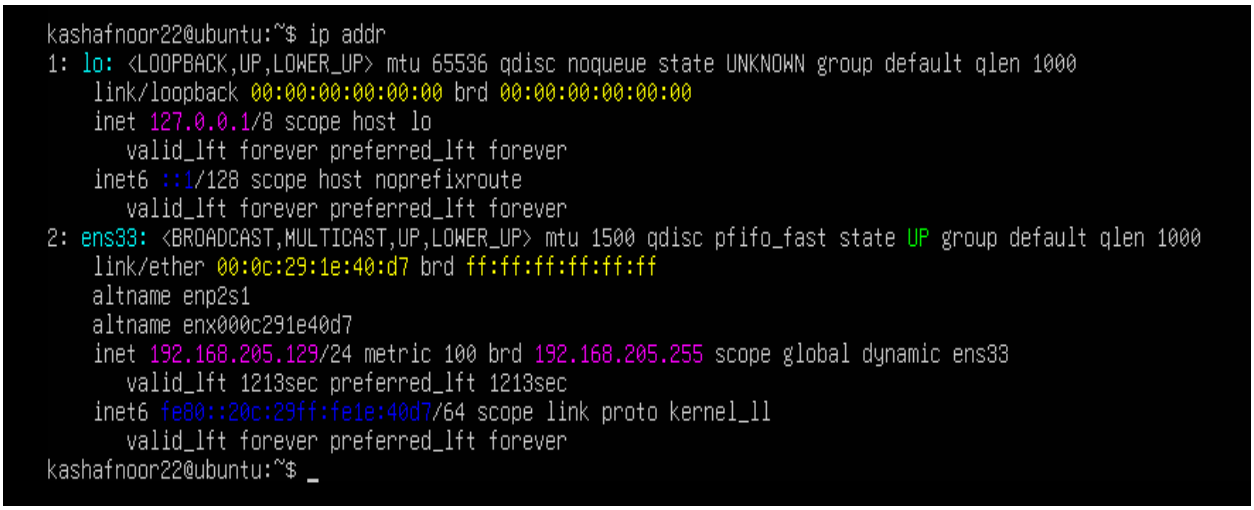
## Task 16: Find the Ubuntu Server IP Address

### Evidence 16.1: Execution of ip addr Command



Caption: Execution of ip addr command in local server terminal.

### Evidence 16.2: Active IPv4 Address Identification

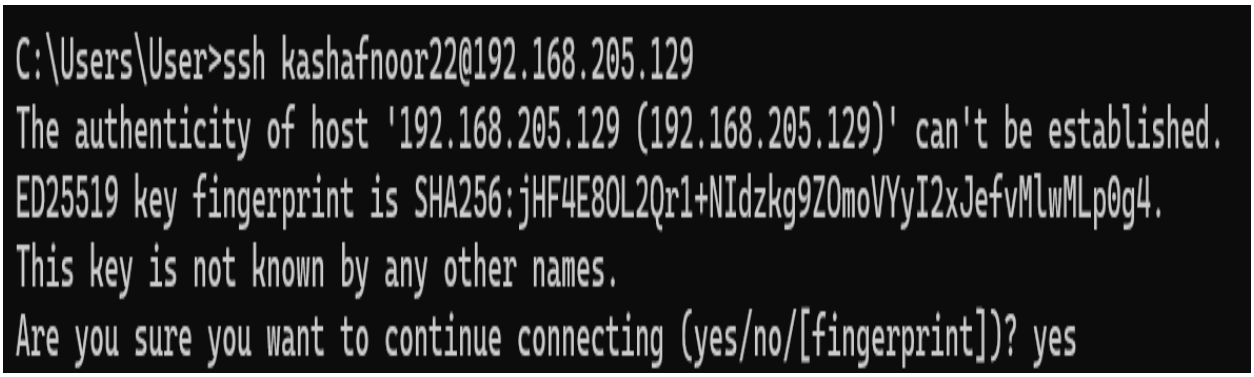



Caption: Terminal output displaying interface ens33 IPv4 address 192.168.201.129 alongside terminal prompt.

## Task 17: Connect to Ubuntu Server from Windows

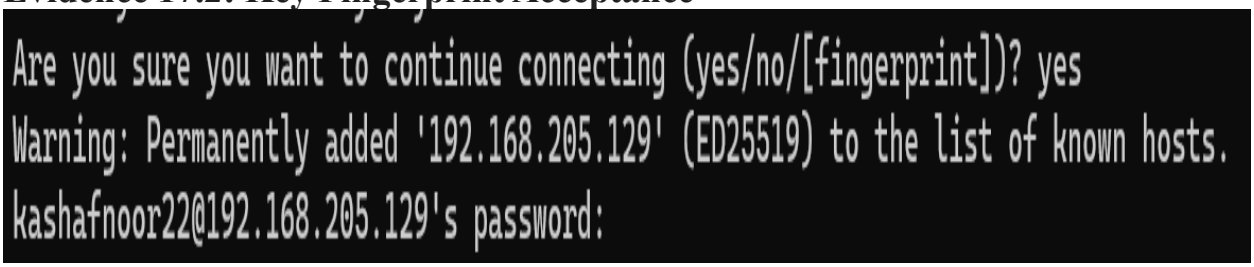
Remote administration setup using SSH protocol from Windows Host Command Prompt.

### Evidence 17.1: SSH Command Execution




Caption: Windows CMD executing command ssh laibaamjad065@192.168.201.129.

### Evidence 17.2: Key Fingerprint Acceptance




Caption: Host key authenticity prompt accepted by typing 'yes' to store ED25519 fingerprint.

## Evidence 17.3: Remote SSH Session Established

```
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Sep 15 04:27:58 PM UTC 2026

  System load:  0.23              Processes:              225
  Usage of /:   51.1% of 9.75GB   Users logged in:        0
  Memory usage: 26%               IPv4 address for ens33: 192.168.205.129
  Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

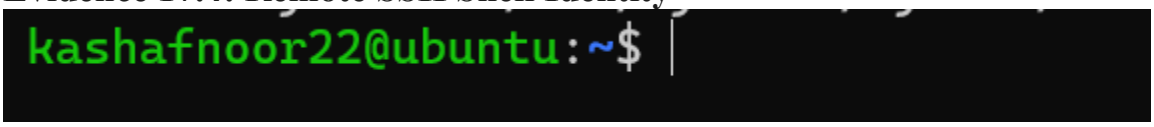
11 updates can be applied immediately.
3 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


kashafnoor22@ubuntu:~$ sudo apt update
```

Caption: Remote session banner display following password verification.

## Evidence 17.4: Remote SSH Shell Identity




Caption: Remote terminal prompt verification displaying laibaamjad065@ubuntu:~$ over SSH.